Supplementary Material:
Deployment Calibration Trace for
The Price of Fairness, Complexity, and Incentives in No-Numeraire Combinatorial Trade-Intent Auctions

This anonymized supplement provides the deployment summary statistics used in Section 7 of the submission, so that the reported figures can be verified without de-anonymizing the authors. The underlying per-auction records and the reproduction scripts are included in the accompanying archive. We reiterate the scope stated in the main text: these are **observable deployment diagnostics, not estimates of the structural complementarity factor** g. The trace records winner and runner-up solution scores and order-level notionals, but not the full vector of per-leg standalone valuations required to compute $V(S)/\sum_j V(\{j\})$; the worst-case results of the submission hold for any g.

S1. Trace scale

Quantity	Value
Post-adoption solver competitions	802,816
Observation window	8 months
Covered notional	\$54.69 B
Visible accounting gap (aggregate)	\$5.13 M
Visible accounting gap (relative)	$\approx$ 0.94 bps

Table 1: Scale of the deployment trace. The visible accounting gap is the aggregate difference between the winning solution's score and the best non-winning visible score on the sanity-filtered USD subsample.

S2. Visible coverage / spread distribution

For each auction, let Score_1 be the winning solution's score and $\text{Score}_2^{\text{other}}$ the best positive score submitted by a different solver. Define the observable

$$\widehat{\alpha}^{\text{obs}} = \frac{\text{Score}_2^{\text{other}}}{\text{Score}_1} \in [0,1], \qquad \widehat{G}^{\text{obs}} = 1/\widehat{\alpha}^{\text{obs}} \geq 1,$$

with $\widehat{G}^{\text{obs}} = \infty$ when $\widehat{\alpha}^{\text{obs}} = 0$. This is an *auction-level* visible coverage; it is a proxy for, and weaker than, the per-leg coverage parameter of the submission's coverage law, which would require per-leg standalone benchmarks.

Event	Share of auctions
Visible spread $\widehat{G}^{\mathrm{obs}} > 2$ (all orders)	25.3%
Visible spread $\widehat{G}^{\mathrm{obs}} > 4$ (all orders)	20.2%
Visible spread $\widehat{G}^{\mathrm{obs}} > 2$ (small-notional orders)	63.1%

Table 2: Tail of the visible-spread distribution. A non-trivial spread is the common case rather than a corner, especially for small orders.

S3. Benchmark robustness

Under a naive local hard-reserve benchmark (reserve computed from a short local window with no inheritance or smoothing), we replay a single low-quote injection per cell-window and record whether it would move the published benchmark enough to admit an otherwise-filtered settlement.

Benchmark variant	Vulnerable cell-windows (auction-weighted)
Local hard reserve (naive)	97.1%

Table 3: Fragility of a naive benchmark to single-injection contamination, motivating the delayed / quantile / rate-limited / parent-cell-inherited benchmarks analyzed in the submission's contamination game. The replay parameters and robust-reserve configuration are in the accompanying scripts.

S4. Reproducibility note

All figures below are computed from an anonymized deployment trace of post-CIP-67 solver competitions. Identifying metadata is removed, and the computation scripts are included in the accompanying anonymized archive.

Sample. The main sample contains 802,816 mainnet solver competitions from June 1, 2025 through January 31, 2026. The parser retains protocol-valid, non-filtered, positive-score solver submissions. Full-sample quantities such as visible spread and effective solver count use all parsed auctions. USD-denominated quantities use smaller denominators: 680,839 score-to-USD observations, 493,653 notional-covered observations, and 318,469 sanity-filtered observations with volume_usd $\in [1, 10^7]$. These denominator changes are intentional coverage restrictions, not inconsistencies.

Score variables. For each competition, Score_1 is the winning solution score. The reference score $\mathrm{Score}_2^{\mathrm{other}}$ is the best positive score submitted by a solver different from the winner. The parser groups by solver identity, removes all candidate solutions submitted by the winning solver, and then takes the maximum remaining positive valid score; if none exists, the reference score is zero. This best-non-winning-solver construction differs from a raw second-solution parser on 11.8% of auctions, and 15.2% of retained auctions have no positive non-winning solver score.

Visible spread. The visible spread is

$$\widehat{G}^{\mathrm{obs}} = \frac{\mathrm{Score}_1}{\mathrm{Score}_2^{\mathrm{other}}},$$

with infinite spread when the best non-winning solver score is zero. The reported tail shares are computed on the full 802,816-auction sample: $\widehat{G}^{\mathrm{obs}} > 2$ in 25.3% of auctions and $\widehat{G}^{\mathrm{obs}} > 4$ in 20.2%. The small-notional segment is the bottom size bucket produced by the score-winner quantile rule $(0, 0.25, 0.75, 0.90, 1.0)$; it contains 200,704 auctions and has $\widehat{G}^{\mathrm{obs}} > 2$ in 63.1%.

Accounting gap. The score difference is converted to dollars using ETH/USD prices:

$$\mathrm{ScoreDiffUSD} = \frac{\mathrm{Score}_1 - \mathrm{Score}_2^{\mathrm{other}}}{10^{18}} \cdot \mathrm{ETHUSD}.$$

The reported \$5.13M aggregate winsorizes this dollar gap at the first and ninety-ninth percentiles and sums over the sanity-filtered denominator above. The \$54.69B covered notional is the sell-token notional over the same denominator, and the 0.94 bps figure is 10^4 times aggregate gap divided by covered notional.

Reserve-vulnerability diagnostic. The benchmark diagnostic uses cell-windows from the reserve-construction pipeline. With baseline quantile $\tau = 0.25$, a cell-window is marked vulnerable if the maximum eligible-sample share of any single solver is at least τ, meaning that a single solver could potentially influence the local reserve quantile if its samples were contaminated. The diagnostic contains 91,390 cell-windows; 99.6% are vulnerable unweighted, and 97.1% are vulnerable after auction-weighting. The corresponding design implication is to inherit from parent cells or use audit-only/soft reserves when local support is sparse.

Scripts. The archive includes the parser and metric construction scripts `build_dataset.py`, `add_usd_prices.py`, and `fragility.py`; validation scripts including `sample_flow.py`, `usd_missingness.py`, `reference_score_validation.py`, and the reserve-vulnerability diagnostic; and table-generation scripts producing the summary CSV and LaTeX tables. Large per-auction parquet files are represented in the submission archive by anonymized processed records or deterministic CSV summaries, depending on upload-size constraints.